INTEL CORPORATION

Enterprise Cloud Analytics (ECA)

&

AI Enablement Architecture

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Prepared by: Sajiv Francis, Don Meyers

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1. Executive Summary

Purpose

This document defines the **Enterprise Cloud Analytics (ECA) & AI Enablement Architecture** for Intel's enterprise data and artificial intelligence capabilities. The architecture provides a unified, governed approach to data ingestion, transformation, serving, and AI-powered consumption.

Scope

The architecture encompasses:

- **Data Platform (ECA)**: End-to-end data flow from SAP source systems through Azure Data Lake Storage (ADLS), Databricks Lakehouse, and Snowflake serving layer
- **AI Platform – Three Architecture Patterns**: Intel's AI enablement is delivered through three complementary architecture patterns, each optimized for specific use case domains:
  - **SAP Joule** – SAP-native AI copilot for transactional, navigational, informational, and analytical interactions within SAP applications (Ariba, Concur, IBP, S/4HANA, etc.)
  - **ECA / Azure Copilot Studio** – Enterprise analytics AI for cross-domain reasoning, agentic workflows, and GenAI/RAG over curated ECA data products (Snowflake, Databricks)
  - **iGPT** – Intel's internal general-purpose Generative AI platform for conversational AI, custom assistant creation, marketplace sharing, and document-grounded knowledge retrieval (igpt.intel.com)
- **Governance**: Comprehensive controls spanning identity, authorization, audit, and compliance across all three patterns

Key Architectural Principles

Principle	Description
Single Source of Truth	ECA serves as the authoritative enterprise data platform
Medallion Architecture	Bronze → Silver → Gold data refinement pattern
Dual Landing Patterns	Standard (batch) and Speed (NRT/RT) data ingestion
AI-Ready Data Products	Curated, governed data products for BI and AI consumption
Governed AI Execution	Tool lifecycle with discover, authorize, execute, audit stages

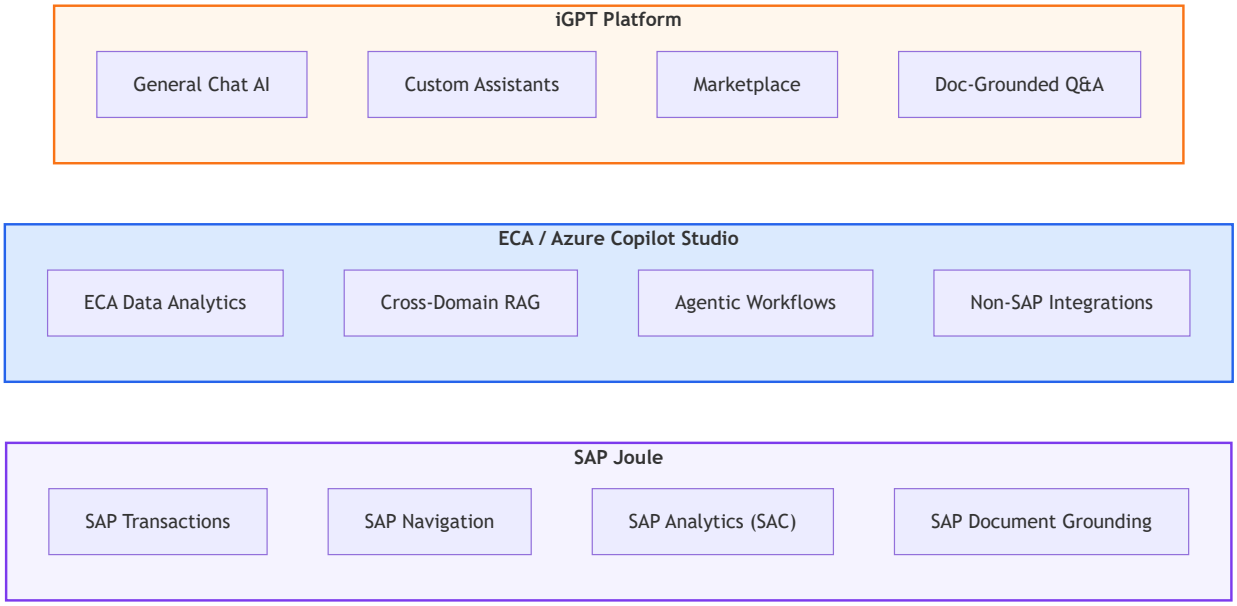
Business Outcomes

- **Finance Reporting Excellence**: Power BI (DARC) delivers trusted finance analytics
- **Near Real-Time Insights**: Speed layer enables sub-minute operational visibility
- **AI-Powered Automation**: Agents and GenAI capabilities augment business processes
- **Governed Self-Service**: Controlled access to enterprise data products

2. AI Architecture Patterns & Routing Guide

Intel's AI enablement strategy is delivered through **three complementary architecture patterns**. Each pattern is purpose-built for a distinct class of use cases with its own runtime, identity model, and governance boundaries. This section provides the decision framework for routing users and use cases to the appropriate architecture.

Three Architecture Patterns at a Glance



Architecture Pattern Comparison

Dimension	SAP Joule	ECA / Azure Copilot Studio	iGPT
Primary Domain	SAP application intelligence	Enterprise data & non-SAP AI	General-purpose conversational AI
Entry Point	Joule Assistant (Work Zone / LoB UIs)	Copilot Agent UX (Teams / MS365)	igpt.intel.com/chat
Build Tool	Joule Studio (SAP Build on BTP)	Copilot Studio (Microsoft)	No-Code Assistant Builder
LLM Provider	SAP AI Core (Generative AI Hub)	Azure OpenAI	LLM Gateway (GPT-4, GPT-3.5, others)
RAG Mechanism	SAP Document Grounding / RAGe	Azure AI Search (vector + hybrid)	iGPT RAG Engine (file upload + pipelines)
Identity	SAP Cloud Identity Services (IAS/IPS)	Microsoft Entra ID	Intel SSO / Azure AD
Governance	BTP RBAC, LoB RBAC, Response Filtering	RBAC/ABAC, Guardrails, Content Safety	Private-by-default, AGS entitlements
Tool Integration	Skills (required for SAP), MCP (optional)	MCP/n8n workflows	None (standalone)
Data Sources	SAP Cloud LoB backends, SAP on-prem (via Cloud Connector)	ECA (Snowflake/Databricks), external APIs	Uploaded documents (PDF/DOCX/TXT), data pipelines
Sharing Model	Role-based (BTP + LoB RBAC)	Role + attribute-based	Private / Public / Secure Groups (Marketplace)
Key Strength	Transactional correctness within SAP	Cross-domain analytics & agentic reasoning	Rapid self-service AI for any Intel employee

Pattern Selection & Detailed Guidance

For comprehensive use case routing, decision frameworks, capability comparisons, hybrid scenarios, and governance-by-pattern analysis, refer to the **AI Architecture Selection Guide**:

> 📄 **AI Architecture Selection Guide** — `AI-Architecture-Selection-Guide.md`

The Selection Guide provides:

- Quick selection table mapping use cases to patterns

- Decision tree flowchart for ambiguous scenarios
- Detailed use case catalog (30+ use cases across all three patterns)
- Hybrid and multi-pattern integration guidance
- Governance and compliance comparison
- Getting started instructions for each pattern

Architecture Pattern References

Pattern	Reference Document	Location
SAP Joule	SAP Joule Architecture Overview	`Current AI Architecture Options/SAP-Joule-Architecture-Overview.md`
ECA / Azure	This document (Sections 4–16)	`ECA_AI_Enablement_Architecture_DocumentV1.md`
iGPT	iGPT Architecture Overview	`Current AI Architecture Options/iGPT-Architecture-Overview.md`
Selection Guide	AI Architecture Selection Guide	`AI-Architecture-Selection-Guide.md`

3. Key Capability Patterns

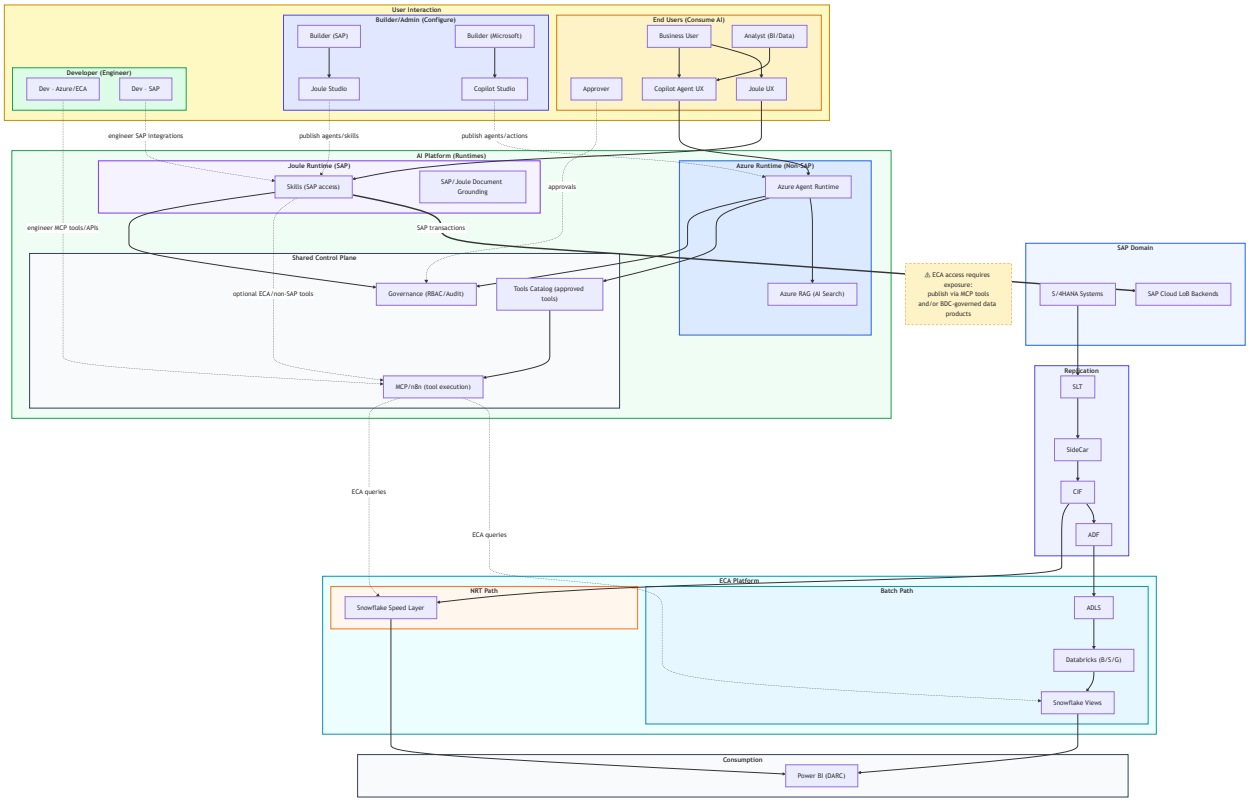
Pattern	Description	ECA Support
Enterprise Data Products	Curated, governed data assets (Finance, Master Data, Supply Chain, KPI) for BI and AI consumption	✅ Primary
Medallion Architecture	Bronze → Silver → Gold data refinement pipeline via Databricks Lakehouse	✅ Primary
RAG (Retrieval-Augmented Generation)	LLM grounding with enterprise data products and documents via Azure AI Search	✅ Primary
Agentic AI	Autonomous agents using governed ECA data tools for analytics and multi-step actions	✅ Primary
Dual Landing Patterns	Standard (batch via ADF) and Speed (NRT/RT via CIF direct) data ingestion	✅ Primary
Complementary Runtime	Joule for SAP transactions, Copilot Studio for ECA analytics — no overlap	✅ Primary
Tool-Based Execution	Governed tool lifecycle: Discover → Authorize → Execute → Audit	✅ Supported
Cross-Domain Analytics	Cross-source reasoning across finance, supply chain, and master data products	✅ Supported

4. Architecture Overview (L0)

L0 – End-to-End Overview: ECA & AI Enablement Architecture

This authoritative overview depicts the complete ECA & AI Enablement architecture, illustrating:

- **SAP Sources**: S/4HANA systems (CFIN, IP/Corp, IF) as primary data sources
- **Replication & Ingestion**: SLT replication through SideCar HANA and CIF control plane
- **ECA Platform**: ADLS landing, Databricks transformation, Snowflake serving
- **AI Platform**: Complementary runtimes (Joule Studio for SAP, Copilot Studio for Azure), MCP/n8n tool orchestration
- **Consumption**: Power BI (DARC) for finance reporting, AI agents for automation



User Interaction Model

End users consume AI through Joule UX (SAP-first assistant, embedded in SAP applications) or Copilot Agent UX (ECA/non-SAP-first, accessible via Teams/MS365). **Builder/Admin users configure and publish** agents, skills, and actions via Joule Studio (SAP) or Copilot Studio (Microsoft)—they do not write code but assemble and govern agent capabilities. **Developers (Code) engineer the underlying runtime and tooling layer**, including MCP tools, ECA query services, SAP integration APIs, and custom connectors. ECA-replicated data is **not native to Joule**; Joule agents that require ECA access must invoke governed MCP tools or BDC-shared data products—there is no implicit or direct query path from Joule to Snowflake/Databricks. Both runtimes invoke the **same governed MCP/n8n tool execution layer** under unified governance controls—end users never interact directly with builder or developer tools.

Runtime Responsibilities and Shared Tool Execution

Joule Runtime is SAP-first: it provides transactional correctness and business-rule fidelity by operating directly on SAP Cloud LoB backends via Skills. **Azure Runtime** is ECA/non-SAP-first: it excels at analytics, cross-domain reasoning, and multi-step agentic workflows over curated enterprise data. Both runtimes may invoke the **same governed tool execution layer (MCP/n8n)** for actions and queries. Joule's access to ECA data is **optional and requires explicit exposure**—ECA data products must be published as MCP tools and/or shared via BDC-governed data products before Joule agents can consume them. This ensures ECA data is never implicitly visible to Joule; all access is governed, auditable, and intentional. **SAP/Joule and Azure maintain separate grounding stacks (SAP-managed grounding vs Azure RAG retrieval); cross-runtime access occurs via governed tool calls, not shared storage.**

Recommended Runtime Model: Complementary Joule + Azure AI

This architecture adopts a **complementary runtime model** where Joule Studio (SAP) and Copilot Studio (Azure) each handle their optimal workloads:

- **Joule Studio is the system of interaction for SAP-native intelligence**, optimized for:
  - SAP Cloud LoB application backends (direct transactional access)
  - SAP transactions, postings, and embedded AI features
  - SAP business semantics, authorization context, and security
- **Copilot Studio is the system of intelligence for enterprise analytics and non-SAP AI**, optimized for:
  - ECA data products (Snowflake views, Databricks Gold)
  - GenAI / RAG over curated, cross-domain enterprise data
  - Agentic workflows across non-SAP systems and external APIs
- **Data replicated from S/4HANA into ECA** becomes analytically optimized, cross-domain enterprise data—intentionally handled by Copilot Studio, not Joule Studio.
- **This separation improves:**
  - Accuracy of SAP transactional AI (operates on live SAP context)
  - Quality of analytical and reasoning-heavy AI (operates on curated data)
  - Governance clarity, scalability, and cost control
- **The two runtimes interoperate** through governed interfaces (ECA tools via MCP, BDC-shared data products)—not direct database access.

Key Architecture Characteristics

Characteristic	Description
Modularity	Each lane operates independently with defined interfaces
Scalability	Databricks and Snowflake provide elastic compute
Governance	Centralized policy enforcement via guardrails and audit
Flexibility	Dual landing patterns support batch and real-time use cases

5. Data Platform Architecture (L1-A)

L1-A – Data Platform Flow: ECA Boundary + CIF Landing Patterns

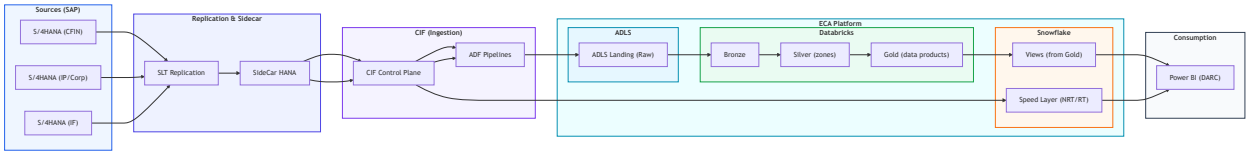
This detailed view illustrates the source-to-target data flow with explicit ECA boundary definition. The Core Ingestion Framework (CIF) supports two landing patterns:

Standard Landing Pattern

- **Path**: ADF → ADLS → Databricks (B/S/G) → Snowflake Views
- **Latency**: Batch (minutes to hours)
- **Use Case**: Historical reporting, analytics, ML training

Speed Landing Pattern

- **Path**: SideCar → CIF → Snowflake Speed Layer (bypasses ADLS/ADF/DBX)
- **Latency**: Near real-time (sub-minute)
- **Use Case**: Operational dashboards, real-time monitoring, agent tools



ECA Boundary Definition

The **Enterprise Cloud Analytics (ECA)** platform boundary encompasses:

Component	Technology	Purpose
Landing Zone	Azure Data Lake Storage (ADLS)	Raw data landing from ADF pipelines
Lakehouse	Databricks	Medallion transformation (Bronze/Silver/Gold)
Serving Layer	Snowflake	Enterprise views and speed layer

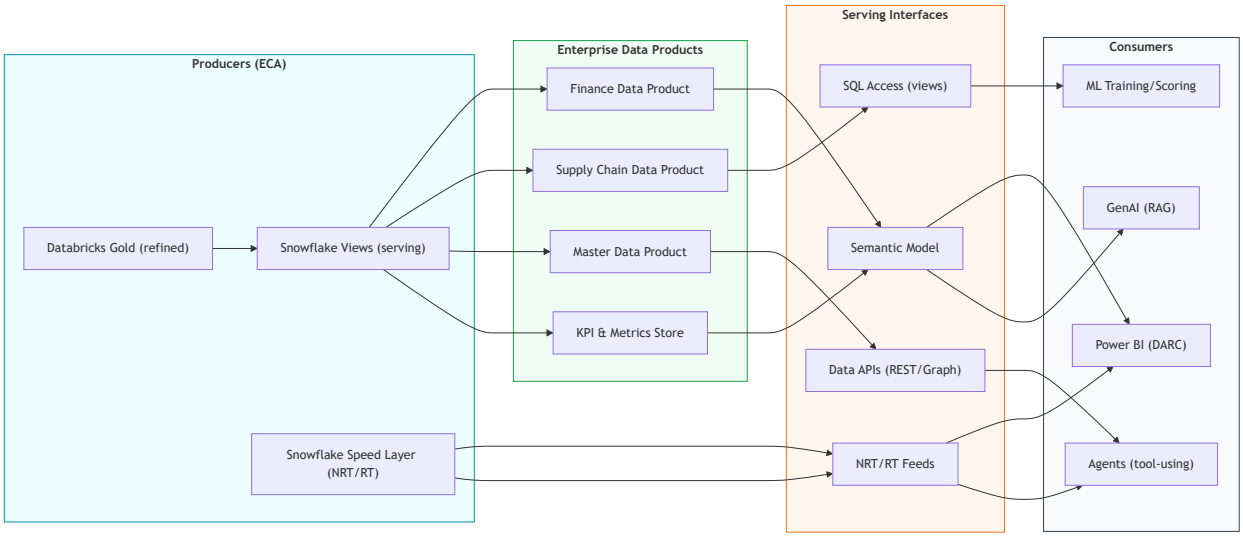
Landing Pattern Comparison

Attribute	Standard Landing	Speed Landing
Entry Point	ADF Pipelines	CIF Direct
Storage	ADLS → Databricks → Snowflake	Snowflake Speed Layer
Transformation	Full medallion (B/S/G)	Minimal (schema mapping)
Latency	Minutes to hours	Sub-minute
Data Freshness	T+1 to T+0 (batch)	Near real-time
Use Cases	Reporting, analytics, ML	Operational, monitoring

6. Data Products & Serving (L1-B)

L1-B – Data Products & Serving: AI-Ready Outputs

This view emphasizes **enterprise data products** as first-class artifacts and illustrates how both BI and AI consumers access governed data through standardized serving interfaces.



Enterprise Data Products Catalog

Data Product	Description	Source	Primary Consumer
Finance Data Product	GL, AP, AR, cost center hierarchies	CFIN	Power BI (DARC)
Master Data Product	Vendor, customer, material masters	IP/Corp	Agents, APIs
Supply Chain Data Product	Inventory, orders, logistics	IF	ML Training
KPI & Metrics Store	Standard definitions, calculations	All	GenAI (RAG), DARC

Serving Interface Specifications

Interface	Protocol	Latency	Access Pattern
Semantic Model	Power BI DirectQuery	Seconds	Interactive BI
SQL Access	Snowflake SQL	Seconds	Ad-hoc queries
Data APIs	REST/GraphQL	Milliseconds	Application integration
NRT/RT Feeds	Snowflake Streams	Sub-minute	Real-time dashboards

ECA AI Integration (Exact Components)

To operationalize AI capabilities on ECA data products, the following components are required:

AI-Ready Data Product Contracts

Each data product must include:

- **Metadata**: Owner, data classification, freshness SLA, allowed columns, row filters
- **Cost Guardrails**: Query limits, compute budgets, rate policies
- **Semantic References**: Links to KPI definitions, business glossary terms

Supporting Stores and Models

Component	Purpose	Integration Point
KPI & Metrics Store	Standard metric definitions	Agent-friendly retrieval via semantic search
Semantic Model	Measures, hierarchies, relationships	Natural language to SQL translation
Business Glossary	Term definitions, synonyms	Context enrichment for RAG grounding

Governed ECA Data Tools

Tool	Data Source	Use Case
Snowflake Views Query Tool	Gold views (batch)	Standard analytics queries, reporting
Snowflake Speed Layer Query Tool	NRT/RT data	Operational dashboards, real-time KPIs
Databricks Query Tool	Gold (primary), Silver (analytics)	ML feature retrieval, complex analytics
Metrics/KPI Tool	KPI Store	Semantic retrieval of standard definitions
Data Catalog/Lineage Tool	Unity Catalog / Snowflake Metadata	Field meaning, provenance, data lineage

Data-to-Text Context Builder

For RAG grounding, structured query results must be transformed:

- Convert tabular results to compact context blocks
- Include provenance IDs for citation/card generation
- Apply row limits and truncation rules
- Format: `{source: "view_name", query_id: "uuid", rows: [...], provenance: "..."}`

AI Tool Controls

Control	Purpose	Default
Query Allowlists	Restrict schemas/views/columns	By role
Max Rows	Limit result set size	1000
Timeout (sec)	Prevent long-running queries	30
Warehouse Policies	Compute tier assignment	X-Small for AI
Rate Limiting	Requests per minute	60/min
ABAC Enforcement	Attribute-based row filtering	Enabled
Security Trimming	Column masking by classification	Enabled

Consumer Access Matrix

This matrix defines **which AI/BI consumers can access which data products** and at what level. It serves as a governance blueprint that drives both RBAC (role-based) and ABAC (attribute-based) access control configurations.

Consumer	Finance	Master Data	Supply Chain	KPI Store
Power BI (DARC)	✅ Primary	✅ Reference	✅ Reference	✅ Primary
ML Training	⬜ Limited	✅ Features	✅ Primary	⬜ Limited
GenAI (RAG)	✅ Context	✅ Context	⬜ Limited	✅ Primary
Agents	✅ Query	✅ Primary	✅ Query	✅ Query

Access Level Definitions

Symbol	Level	Description	Governance Implication
✅ Primary	Full access	Consumer's main data source; optimized for their use case	Full RBAC grant; no row/column restrictions
✅ Reference	Lookup access	Can join/reference for enrichment, not primary dataset	Read-only RBAC; may have column restrictions
✅ Context	Grounding	Used to enrich AI responses with business context	Read-only; ABAC filters sensitive rows
✅ Query	On-demand	Can query as needed for specific tasks	Query-level RBAC; rate limits apply
✅ Features	ML input	Used as input features for machine learning models	Read-only; anonymization may apply
⬜ Limited	Restricted	Constrained due to data sensitivity or compliance	ABAC row filtering; column masking; audit required

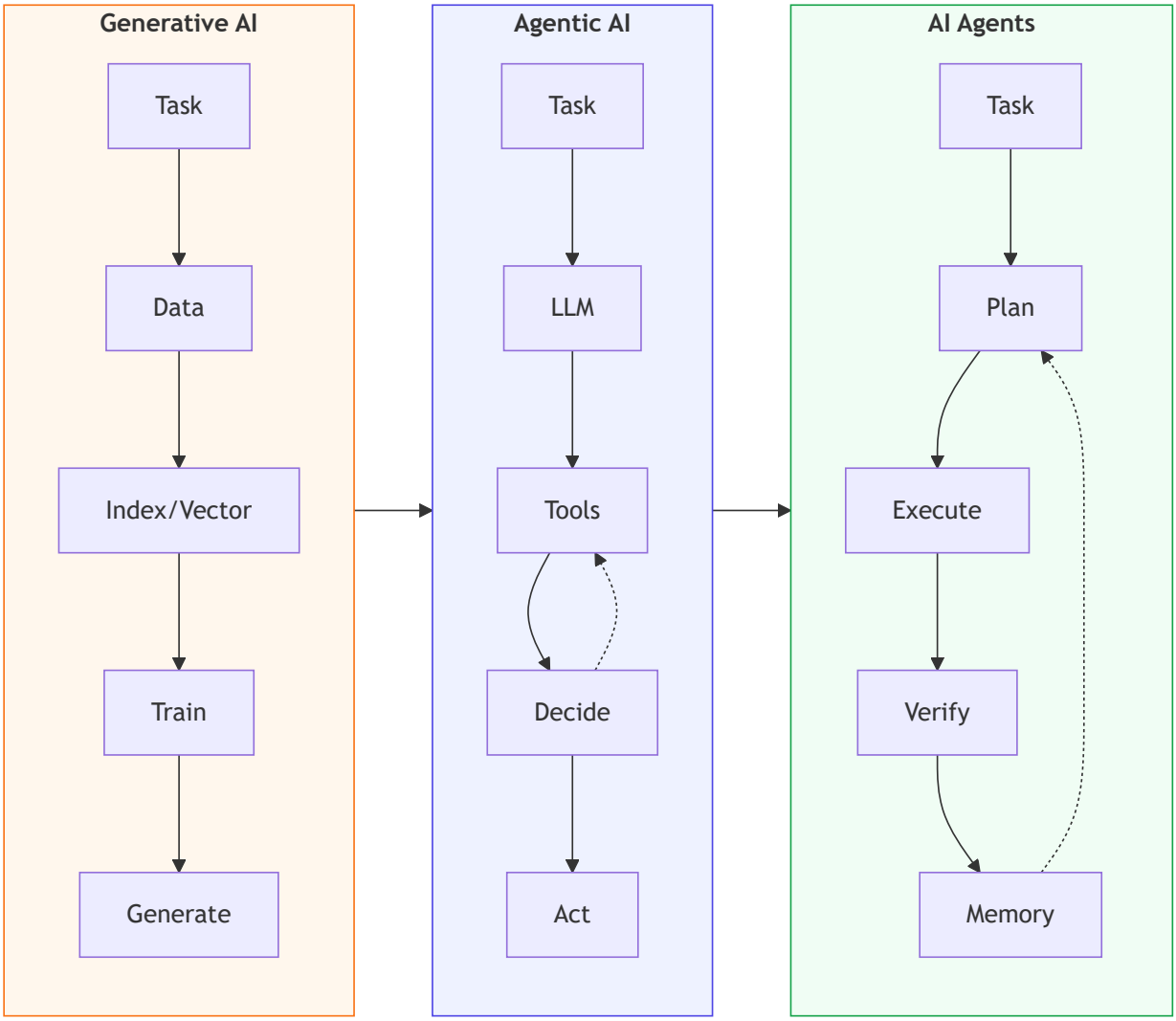
How This Matrix Drives Access Control

- **RBAC (Role-Based)**: Consumer type (Power BI, ML, GenAI, Agents) determines the role and base permissions
- **ABAC (Attribute-Based)**: Data classification, user attributes, and context determine row-level filtering and column masking
- **Combined enforcement**: A consumer must have both the correct role (RBAC) AND satisfy attribute conditions (ABAC) to access data

7. AI Build Modes Comparison (L1-C)

L1-C – AI Build Modes: GenAI vs Agentic AI vs AI Agents

This comparison illustrates the spectrum of AI capabilities, from task-specific generative AI through increasingly autonomous agentic patterns.



AI Mode Comparison Matrix

Characteristic	Generative AI	Agentic AI	AI Agents
Primary Pattern	Retrieve + Generate	Plan + Execute + Iterate	Multi-step + Memory + Adapt
Autonomy Level	Low (task-specific)	Medium (self-directing)	High (goal-oriented)
Memory	None (stateless)	Short-term (session)	Long-term (persistent)
Tool Usage	Limited (retrieval)	Extensive (API calls)	Comprehensive (actions)
Learning	Pre-trained + RAG	In-context learning	Adaptive over time
Human Oversight	Output review	Checkpoint approval	Action approval gates

Use Case Alignment

Use Case	Recommended Mode	Rationale
Policy Q&A chatbot	Generative AI (RAG)	Retrieve documents, generate answers
Data analysis assistant	Agentic AI	Query data, iterate on insights
Automated workflow executor	AI Agents	Multi-step processes with approvals
Report generation	Generative AI	Template-based generation
Exception handling	AI Agents	Decision-making with memory
Code assistance	Agentic AI	Tool integration, iterative refinement

Governance Requirements by Mode

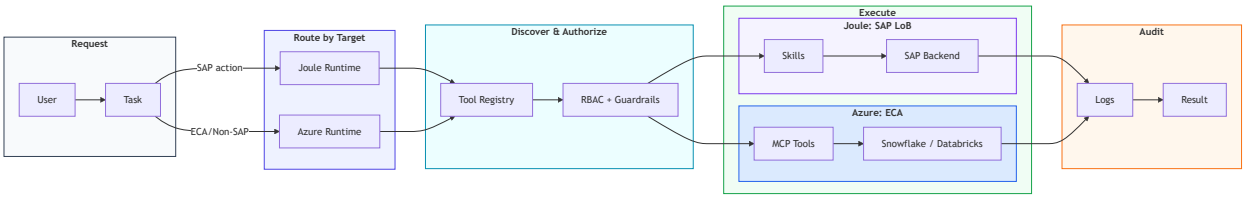
Control	Generative AI	Agentic AI	AI Agents
Input Validation	✅ Required	✅ Required	✅ Required
Output Filtering	✅ Required	✅ Required	✅ Required
Action Approval	⬜ N/A	✅ Recommended	✅ Required
Audit Trail	✅ Required	✅ Required	✅ Required
Memory Governance	⬜ N/A	⬜ Optional	✅ Required

Legend: ✅ Required = Must implement | ✅ Recommended = Strongly advised | ⬜ Optional = Implement if needed | ⬜ N/A = Not applicable to this mode

8. AI Platform Tool Lifecycle (L1-D)

L1-D – AI Platform Tool Lifecycle: Discover → Authorize → Execute → Audit

This diagram shows the end-to-end tool lifecycle used by both Joule Studio (SAP) and Copilot Studio (Azure). The complementary model routes requests to the appropriate runtime based on the target system.



Runtime Routing: When to Use Joule vs Copilot Studio vs iGPT

The architecture routes AI requests to the appropriate runtime based on the target system and interaction pattern. Refer to **Section 2 – AI Architecture Patterns & Routing Guide** for the full decision framework.

Target	Runtime	Access Method
SAP transactions (read/write)	Joule Studio	Skills (required)
SAP embedded features	Joule Studio	Skills (required)
SAP analytics (SAC)	Joule Studio	SAC Just Ask
ECA data (Snowflake/Databricks)	Copilot Studio	MCP tools
Non-SAP APIs	Copilot Studio	MCP/n8n workflows
BDC data products	Joule or Copilot Studio	BDC Connect
General conversational AI	iGPT	igpt.intel.com/chat
Custom assistant creation (no-code)	iGPT	Assistant Builder + Marketplace
Document-grounded Q&A (uploaded files)	iGPT	RAG Engine (file upload)

Key principles:

- Joule Skills are required only for SAP-native access. ECA data—even though it originates from SAP—is accessed via Copilot Studio using governed MCP tools.
- iGPT is the self-service option for Intel employees who need general-purpose AI without SAP or ECA integration requirements.
- Use cases requiring both SAP transactions and ECA analytics should leverage the **Joule + ECA hybrid** pattern (Joule for the transaction, MCP bridge for analytics).

Tool Lifecycle Stages

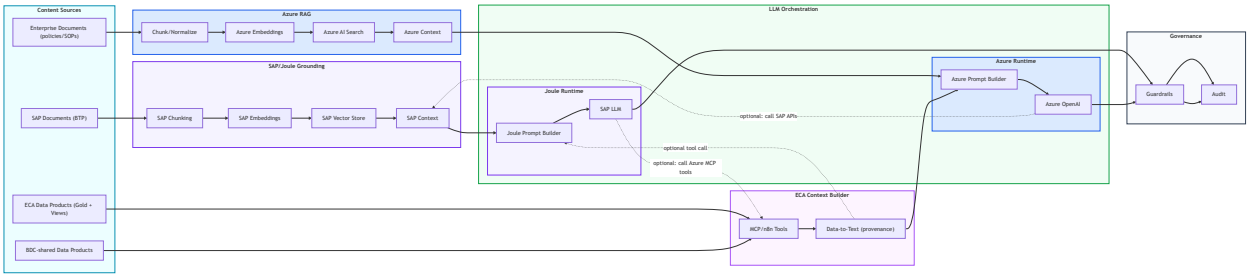
All AI tool calls—whether through Joule Studio or Copilot Studio—follow a governed lifecycle:

Stage	Purpose	Key Controls
Discover	Find relevant tools for the task	Tool registry, capability matching
Authorize	Verify access rights	RBAC/ABAC, guardrails, approval gates
Execute	Invoke the tool	Joule Skills (SAP) or MCP (ECA/non-SAP)
Audit	Record outcomes	Logs, traces, evidence retention

9. GenAI RAG Implementation (L2-D)

L2-D – GenAI RAG Flow Using ECA Data Products

This implementation-level view details the Retrieval-Augmented Generation (RAG) pattern, grounding LLM responses with enterprise data products and documents. **Importantly, SAP/Joule and Azure runtime maintain separate retrieval mechanisms by default.**



RAG Retrieval Separation: SAP/Joule vs Azure

The complementary runtime model extends to RAG retrieval—**SAP/Joule and Azure maintain separate document grounding mechanisms by default**:

- **SAP/Joule uses SAP-managed document grounding**: SAP BTP provides its own RAG infrastructure (chunking, embeddings, vector store) optimized for SAP content, business semantics, and authorization context.
- **Azure runtime uses Azure-native retrieval**: Azure AI Search and vector stores index enterprise documents, ECA data products, and non-SAP content for Azure-based agents.
- **Indexes are separate by default**: Even when source documents overlap, each runtime maintains its own embeddings and retrieval index—there is no implicit shared vector store.
- **ECA structured data is accessed via MCP tools**: Both runtimes can invoke governed MCP tools to query Snowflake views, Speed Layer, and Databricks Gold products for real-time structured context.

- **Cross-runtime retrieval is optional and governed**: Joule can call Azure-side MCP tools for ECA facts; Azure can call SAP APIs when SAP context is needed—but these are explicit, auditable tool calls, not automatic sharing.
- **BDC-shared data products bridge both**: When data products are published via BDC Connect, both runtimes can access them through their respective governed interfaces.
- **This separation improves**: Security (no cross-runtime data leakage), governance (clear audit trails), and optimization (each runtime's RAG tuned for its content type).

RAG Pipeline Components

Input Sources

Source Type	Examples	Refresh Frequency
Enterprise Data Products	Finance metrics, master data, KPIs	Daily (batch)
Enterprise Documents	Policies, SOPs, runbooks, guides	On change
Structured Data	Snowflake views, API responses	Real-time

Retrieval & Grounding Pipeline

Stage	Process	Technology
Chunking	Split documents into semantic units	LangChain, custom splitters
Embedding	Convert chunks to vector representations	Azure OpenAI, enterprise models
Indexing	Store embeddings with metadata	Azure AI Search, Pinecone
Retrieval	Semantic search for relevant context	Vector similarity, hybrid search

LLM Orchestration

Component	Function	Configuration
Prompt Builder	Assemble task + context + constraints	Template library
LLM Runtime	Generate response	GPT-4, Claude, enterprise-approved
Response Composer	Format output with citations	Cards, tables, references

Governance Controls

Control	Implementation	Purpose
Guardrails	Azure AI Content Safety	Filter harmful content
PII Detection	Presidio, custom rules	Protect sensitive data
Citation Tracking	Source attribution	Traceability, trust
Audit Logging	Application Insights	Compliance, debugging

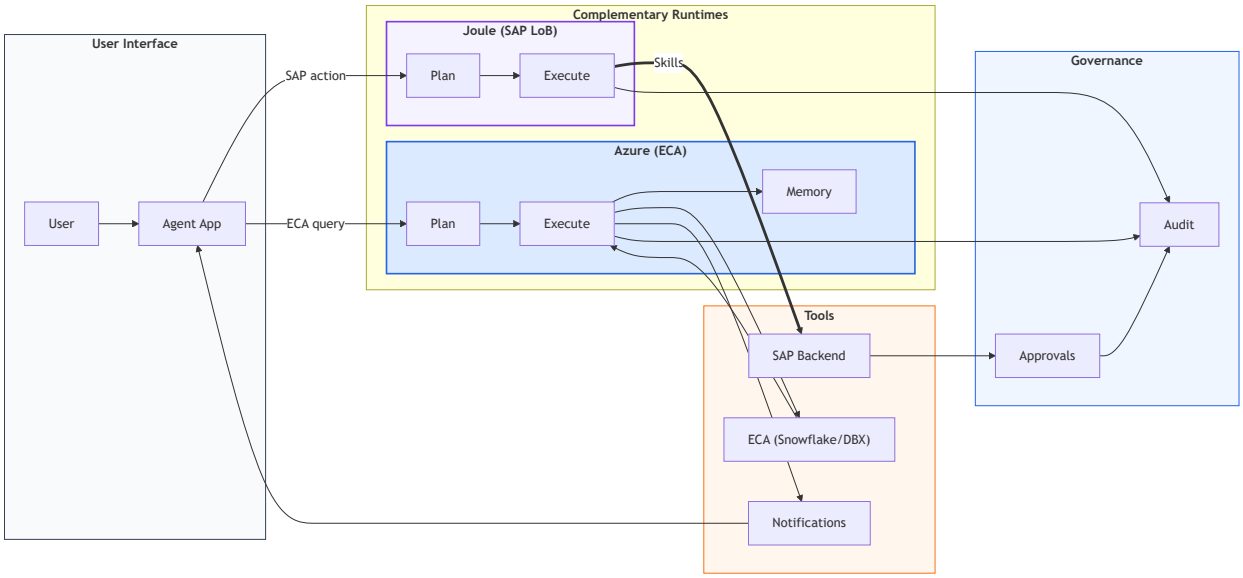
RAG Quality Metrics

Metric	Target	Measurement
Retrieval Precision	> 85%	Relevant chunks / retrieved chunks
Answer Accuracy	> 90%	Correct answers / total answers
Citation Accuracy	> 95%	Valid citations / total citations
Response Latency	< 3 sec	P95 end-to-end time

10. Agent Runtime Implementation (L2-E)

L2-E – Agent Runtime: Complementary Model

This diagram shows the agent runtime architecture in the complementary model, where Joule handles SAP transactions and Azure handles ECA analytics.



Agent Runtime Components

Planner

Function	Description	Implementation
Goal Decomposition	Break complex tasks into steps	LLM reasoning, ReAct pattern
Tool Selection	Match steps to available tools	Tool registry lookup
Dependency Ordering	Sequence steps appropriately	DAG construction
Error Recovery	Handle failures, re-plan	Retry logic, fallbacks

Executor

Function	Description	Implementation
Tool Invocation	Call selected tools with parameters	API calls, MCP protocol
Result Processing	Parse and validate tool outputs	Schema validation
State Management	Track execution progress	Session state, checkpoints
Error Handling	Manage tool failures	Circuit breakers, retries

Memory

Type	Scope	Storage	Retention
Short-term	Session	In-memory	Session duration
Long-term	User/Agent	Database	Configurable
Episodic	Task history	Vector store	Policy-defined

Tool Categories

Data Tools (Read)

Tool	Source	Access Pattern	Latency
Snowflake Views	Curated data products	SQL query	Seconds
Speed Layer	NRT/RT data	Streaming query	Sub-second
Databricks	Bronze/Silver/Gold	Spark SQL	Seconds

Action Tools (Write)

Tool	Target	Risk Level	Approval Required
SAP APIs	Transaction posting	High	Yes (HITL)
Notifications	Teams/Email	Low	No
Ticketing	ServiceNow	Medium	Configurable

Risk Level Definitions: High = Financial/compliance impact, requires HITL | Medium = Operational impact, approval configurable | Low = Minimal impact, auto-approved

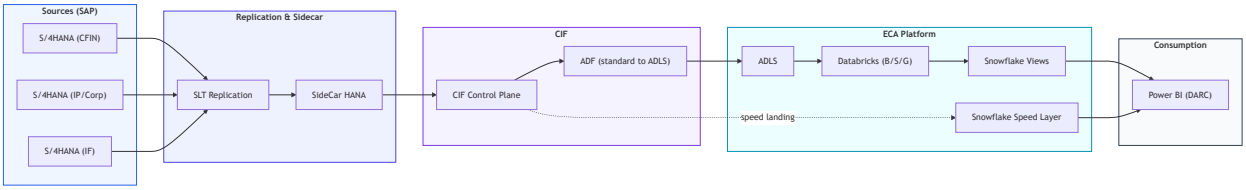
Governance Framework

Control	Scope	Enforcement
RBAC/ABAC	All tool access	Pre-execution check
Approval Gate	High-risk actions	Human approval required
Rate Limiting	API calls	Token bucket
Audit Logging	All operations	Post-execution capture

11. Platform Stage Lanes

Platform Stage Lanes – Simplified View

This simplified lane view provides an executive-friendly overview of the data platform stages, ideal for stakeholder presentations and governance discussions.



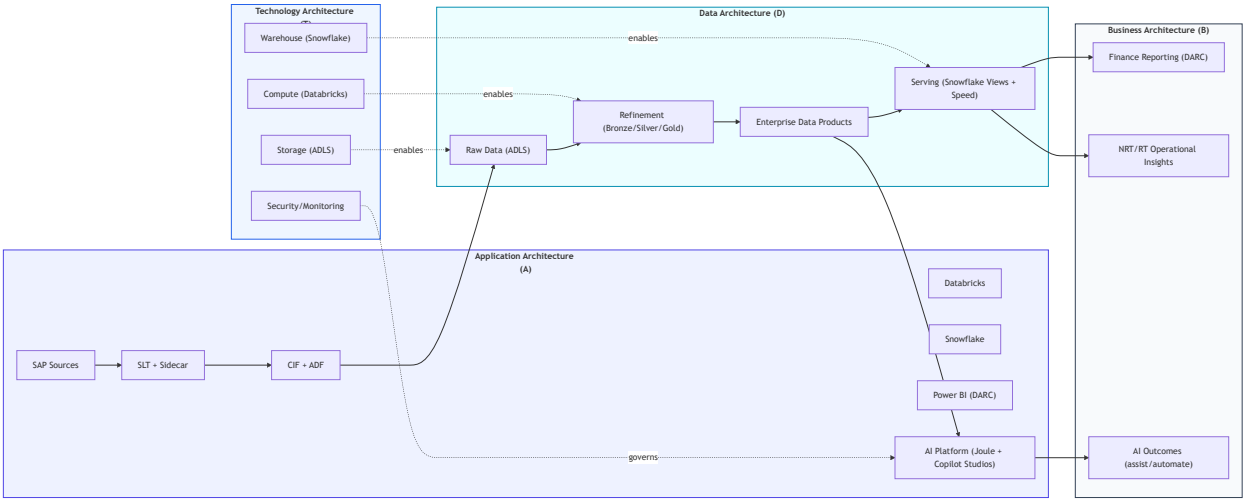
Stage Summary

Stage	Components	Responsibility	SLA
Sources	S/4HANA (CFIN, IP/Corp, IF)	SAP App Team	99.9% availability
Replication	SLT, SideCar HANA	Integration Team	< 5 min replication lag
Ingestion	CIF, ADF Pipelines	Data Engineering	99.5% pipeline success
ECA	ADLS, Databricks, Snowflake	Data Platform Team	99.9% availability
Consumption	Power BI (DARC)	BI Team	< 10 sec query response

12. TOGAF BDAT Crosswalk

TOGAF BDAT Crosswalk – Conceptual Mapping

This conceptual mapping aligns the ECA AI Architecture Overview to TOGAF Business, Data, Application, and Technology (BDAT) domains, supporting enterprise architecture governance.



BDAT Domain Mapping

Business Architecture (B)

Capability	Description	Value Delivered
Finance Reporting	Power BI dashboards via DARC	Trusted financial insights
NRT/RT Insights	Operational monitoring	Real-time decision support
AI Outcomes	Agent-assisted processes	Automation, efficiency

Data Architecture (D)

Layer	Purpose	Quality Standard
Raw Data (ADLS)	Landing zone	Schema-on-read
Bronze	Initial ingestion	Data quality checks
Silver	Cleansed, conformed	Business rules applied
Gold	Curated products	Certified, documented
Serving	Enterprise access	SLA-backed availability

Application Architecture (A)

Application	Role	Integration Pattern
SAP Sources	System of record	CDC replication
SLT + Sidecar	Replication engine	Near real-time
CIF + ADF	Ingestion control	Batch/streaming
Databricks	Transformation	Medallion pipeline
Snowflake	Serving layer	SQL/API access
Power BI	Visualization	DirectQuery
AI Platform	Intelligence layer	Tool/API integration

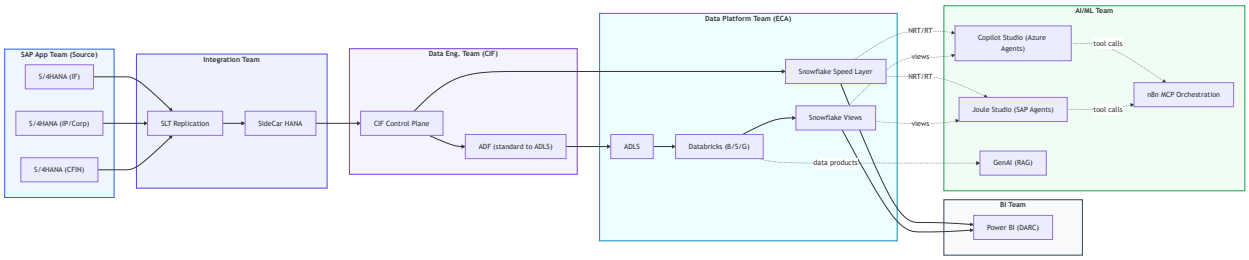
Technology Architecture (T)

Technology	Purpose	Vendor
ADLS	Object storage	Microsoft Azure
Databricks	Lakehouse compute	Databricks
Snowflake	Cloud warehouse	Snowflake
Azure Monitor	Observability	Microsoft Azure
Entra ID	Identity	Microsoft Azure

13. Ownership & Operating Model

Ownership / Operating Model Lanes

This view maps platform components to responsible teams, supporting RACI discussions and operational governance.



RACI Matrix

Component	SAP App	Integration	Data Eng	Data Platform	BI Team	AI/ML
S/4HANA Systems	**R/A**	C	I	I	I	I
SLT Replication	C	**R/A**	C	I	I	I
SideCar HANA	C	**R/A**	C	I	I	I
CIF Control Plane	I	C	**R/A**	C	I	I
ADF Pipelines	I	I	**R/A**	C	I	I
ADLS	I	I	C	**R/A**	I	I
Databricks	I	I	C	**R/A**	I	C
Snowflake Views	I	I	C	**R/A**	C	C
Speed Layer	I	I	C	**R/A**	C	C
Power BI (DARC)	I	I	I	C	**R/A**	I
Joule Studio	I	I	I	C	I	**R/A**
Copilot Studio	I	I	I	C	I	**R/A**
n8n MCP	I	I	I	C	I	**R/A**
GenAI (RAG)	I	I	I	C	I	**R/A**

RACI Legend:

Code	Role	Definition
R/A	Responsible & Accountable	Owns delivery and final decision authority
C	Consulted	Provides input before decisions are made
I	Informed	Notified after decisions are made

Support Model

Tier	Scope	Team	SLA
L1	User issues, access requests	Service Desk	4 hours
L2	Data quality, pipeline failures	Data Engineering	8 hours
L3	Platform issues, performance	Data Platform	24 hours
L4	Architecture, design decisions	Enterprise Architecture	5 days

14. Implementation Phases

Phase 1: Data Platform Foundation

Purpose: Establish core ECA data infrastructure

Deliverable	Description	Status
ADLS Landing Zones	Raw data storage structure	✅ Complete
Databricks Workspace	B/S/G layer configuration	✅ Complete
Snowflake Views	Enterprise serving layer	✅ Complete
CIF Standard Landing	ADF pipeline integration	✅ Complete
DARC Connectivity	Power BI dataset configuration	✅ Complete

Status Legend: ✅ Complete = Delivered and operational | 🔄 In Progress = Active development | 📋 Planned = Scheduled for future phase

L2 Deliverables: Infrastructure architecture, data model design

Phase 2: Speed Layer Enablement

Purpose: Enable near-real-time data access

Deliverable	Description	Status
Snowflake Speed Layer	NRT/RT data landing	✅ Complete
CIF Speed Landing	Direct CIF → Snowflake path	✅ Complete
NRT Dashboard Integration	DARC real-time refresh	🔄 In Progress
Speed Layer Governance	Access controls, monitoring	🔄 In Progress

L2 Deliverables: NRT architecture, latency requirements

Phase 3: Enterprise Data Products

Purpose: Create reusable, governed data products

Deliverable	Description	Status
Finance Data Product	GL, AP, AR, cost centers	✅ Complete
Master Data Product	Vendor, customer, material	🔄 In Progress
Supply Chain Data Product	Inventory, orders, logistics	📋 Planned
KPI & Metrics Store	Standard definitions	📋 Planned
Semantic Layer	Measures, hierarchies	📋 Planned

L2 Deliverables: Data product catalog, semantic layer design

Phase 4: AI Platform Foundation

Purpose: Enable AI/ML consumption of data products

Deliverable	Description	Status
LLM Runtime	Enterprise-approved model hosting	🔄 In Progress
Tool Registry	Catalog of available tools	📋 Planned
Guardrails Framework	Content safety, policy enforcement	📋 Planned
Vector Store	RAG grounding storage	📋 Planned
Observability	Tracing, metrics, logging	📋 Planned

L2 Deliverables: AI platform architecture, security controls

Phase 5: Agent Studio Enablement

Purpose: Enable custom agent development on both platforms

Deliverable	Description	Status
Joule Studio Deployment	SAP agent development platform	📋 Planned
Copilot Studio Deployment	Azure agent development platform	📋 Planned
Skills Library	ECA data tools integration	📋 Planned
n8n MCP Server	Workflow orchestration	📋 Planned
Approval Gates	HITL for high-risk actions	📋 Planned
Agent Templates	Pre-built agent patterns	📋 Planned

L2 Deliverables: Agent architecture, tool lifecycle design

Phase 6: Governance & Operations

Purpose: Establish ongoing operational excellence

Deliverable	Description	Status
Audit Framework	Comprehensive logging, evidence	📋 Planned
RBAC/ABAC Controls	Cross-platform access management	📋 Planned
SLA Monitoring	Alerting, dashboards	📋 Planned
Operating Model	RACI, support procedures	📋 Planned
Runbooks	Operational procedures	📋 Planned

L2 Deliverables: Governance framework, operating model

Implementation Timeline

```
Phase 1 [==========] Complete - Data Platform Foundation
Phase 2 [========  ] 80% - Speed Layer Enablement
Phase 3 [====      ] 40% - Enterprise Data Products
Phase 4 [==        ] 20% - AI Platform Foundation
Phase 5 [          ] 0% - Agent Studio Enablement
Phase 6 [          ] 0% - Governance & Operations
```

15. L2 Architecture Deliverables

15.1 Data Architecture

ADLS Structure

```
/raw/
  ├── cfin/           # CFIN source data
  ├── ipcorp/         # IP/Corp source data
  └── if/             # IF source data
/bronze/
  ├── tables/         # Raw table replicas
  └── metadata/       # Schema, lineage
/silver/
  ├── finance/        # Finance domain
  ├── masterdata/     # Master data domain
  └── supplychain/    # Supply chain domain
/gold/
  ├── dataproducts/   # Certified products
  └── kpi/            # Metrics store
```

Databricks Medallion Layers

Layer	Purpose	Quality Rules	Retention
Bronze	Raw ingestion	Schema validation	90 days
Silver	Cleansed, conformed	Business rules, dedup	1 year
Gold	Certified products	Data quality scores	7 years

Snowflake Organization

Schema	Purpose	Access Pattern
`views`	Enterprise serving	DirectQuery, SQL
`speed`	NRT/RT data	Streaming, real-time
`staging`	Temporary processing	Internal only

15.2 Integration Architecture

Standard Landing Pattern

```
SAP S/4HANA → SLT (CDC) → SideCar HANA → CIF → ADF → ADLS → Databricks → Snowflake
```

Configuration:

- SLT: Real-time CDC with 5-minute latency
- ADF: Hourly batch pipelines
- Databricks: Delta Lake with auto-optimize

Speed Landing Pattern

```
SAP S/4HANA → SLT (CDC) → SideCar HANA → CIF → Snowflake Speed Layer
```

Configuration:

- Direct CIF → Snowflake Snowpipe
- Sub-minute latency
- Schema-on-read with validation

15.3 Security Architecture

Identity & Access

Component	Identity Provider	Access Control
ADLS	Entra ID	RBAC (Storage Blob)
Databricks	Entra ID	Unity Catalog
Snowflake	Entra ID (SAML)	Role-based
Power BI	Entra ID	Row-level security
AI Platform	Entra ID	RBAC + ABAC

Data Classification

Classification	Examples	Controls
Public	Published reports	None
Internal	Operational data	Authentication
Confidential	Financial details	Encryption, audit
Restricted	PII, trade secrets	Encryption, approval, audit

Network Security

Zone	Components	Access
DMZ	API Gateway, WAF	Internet
Application	Power BI, AI Platform	Corporate
Data	ADLS, Databricks, Snowflake	Private endpoints
Management	Monitoring, secrets	Admin only

15.4 AI Platform Architecture

LLM Runtime

Model	Provider	Use Case	Approval
GPT-4	Azure OpenAI	General, RAG	✅ Approved
Claude 3	Anthropic	Complex reasoning	✅ Approved
Llama 3	Meta (self-hosted)	Sensitive data	🔄 Evaluation

Approval Status: ✅ Approved = Cleared for production use | 🔄 Evaluation = Under security/compliance review | ⬜ Pending = Not yet reviewed

Tool Registry

Tools are governed, cataloged interfaces to data and actions. Each tool in the registry includes:

- Identity and description
- Risk level and required approvals
- Access controls and allowed resources
- Rate limits and cost controls

The exact schema is an implementation detail and intentionally out of scope for this architecture document. What matters architecturally is that all tools—whether Joule Skills or MCP workflows—are registered, governed, and auditable.

Joule Studio Connectivity to External (Non-SAP-Managed) Databricks & Snowflake

Joule Studio agents can leverage customer-managed Databricks and Snowflake platforms via **SAP Business Data Cloud (BDC) Connect**, enabling governed, bidirectional, zero-copy data sharing with semantic preservation.

Important: Joule Studio does not typically bypass BDC for grounding; BDC is the unification and governance layer for semantically rich data product sharing. MCP tools can still call external systems for execution, but BDC Connect is the primary mechanism for governed data product sharing and grounding.

Databricks Connectivity via BDC Connect

Mechanism: BDC Connector for Databricks using Delta Sharing (zero-copy)

Capabilities:

- Bidirectional sharing: BDC → Databricks and Databricks → BDC
- Zero-copy data access (no data movement)
- Semantic model preservation across platforms
- Catalog discoverability for AI grounding

Setup Requirements:

- Unity Catalog enabled on Databricks workspace
- Delta Sharing enabled and configured
- Admin privileges: CREATE PROVIDER, CREATE RECIPIENT
- Network connectivity between BDC and Databricks
- Credential configuration in BDC Connect service

Snowflake Connectivity via BDC Connect

Mechanism: BDC Connect service for Snowflake using Snowflake Data Sharing

Capabilities:

- Bidirectional zero-copy sharing
- Semantically rich data products with metadata
- Governance policy enforcement
- Catalog discoverability for AI grounding

Setup Requirements:

- Snowflake account with Data Sharing enabled
- ACCOUNTADMIN or CREATE SHARE privileges
- Network policy allowing BDC connectivity
- Share configuration in BDC Connect service

Limitations & Considerations

Consideration	Description
Latency	Zero-copy sharing adds minimal latency vs. direct JDBC
Governance	All access subject to BDC governance policies
Schema Evolution	Schema changes require re-sync with BDC catalog
Cost	Compute costs incurred at source platform
Security	Cross-platform credentials managed in BDC
Scope	Grounding only; MCP handles tool execution

Azure AI Runtime Components

The Azure runtime in the complementary model handles ECA analytics and non-SAP AI workloads:

Component	Purpose
Agent Runtime	Orchestration, planning, execution
LLM Endpoint	Enterprise-approved model hosting (Azure OpenAI)
Vector Store	RAG grounding, semantic search (Azure AI Search)
MCP/n8n	Tool execution, workflow orchestration
Observability	Tracing, metrics, audit logging

The Azure runtime accesses ECA data via governed MCP tools (Snowflake queries, Databricks queries) and does not require Joule Skills for ECA access.

Agent Governance

Control	Implementation	Trigger
Input validation	Schema check	All requests
Output filtering	Content safety	All responses
Action approval	HITL workflow	High-risk actions
Rate limiting	Token bucket	All API calls
Audit logging	Application Insights	All operations

16. Governance & Operations

Governance Framework

Data Governance

Domain	Owner	Controls
Data Quality	Data Platform Team	Quality scores, validation rules
Data Lineage	Data Engineering	Automated tracking, documentation
Data Catalog	Data Platform Team	Discovery, metadata management
Data Access	Security Team	RBAC, classification enforcement

AI Governance

Domain	Owner	Controls
Model Approval	AI/ML Team	Enterprise model registry
Prompt Safety	AI/ML Team	Guardrails, content filtering
Action Approval	Business Owners	HITL for high-risk actions
Audit & Evidence	Compliance	Comprehensive logging

Operational Procedures

Monitoring & Alerting

Metric	Threshold	Action
Pipeline failure rate	> 5%	Page on-call
Query latency P95	> 10 sec	Investigate
LLM error rate	> 1%	Alert AI team
Approval backlog	> 50	Escalate

Incident Management

Severity	Definition	Response Time	Resolution Time
P1	Platform outage	15 min	4 hours
P2	Major degradation	1 hour	8 hours
P3	Minor issue	4 hours	24 hours
P4	Enhancement request	1 week	Sprint planning

Change Management

Change Type	Approval	Lead Time
Standard (pre-approved)	Auto	Immediate
Normal	CAB	5 days
Emergency	Emergency CAB	2 hours

17. Architectural Flows

Standard Data Landing Flow

```
SAP S/4HANA (CFIN / IP / IF) → SLT (CDC Replication) →
SideCar HANA → CIF Control Plane → ADF Pipelines →
ADLS Landing (Raw) → Databricks Bronze → Silver → Gold →
Snowflake Views → Power BI (DARC)
```

Speed Data Landing Flow

```
SAP S/4HANA (CFIN / IP / IF) → SLT (CDC) → SideCar HANA →
CIF Control Plane → Snowflake Speed Layer (NRT/RT) → Power BI (DARC)

Note: Speed landing bypasses ADLS/Databricks for sub-minute latency.
```

GenAI RAG Retrieval Flow

```
User → Agent App → Azure Runtime (Plan) →
RAG Engine (Azure AI Search) → Vector DB (Semantic Search) →
Context Reranking → Prompt Builder (Task + Context + Constraints) →
Azure OpenAI (LLM) → Guardrails → Response with Citations → User

Note: SAP/Joule maintains separate RAG infrastructure.
Cross-runtime retrieval is optional via governed MCP tool calls.
```

Agentic Query Flow (ECA Data)

```
User → Agent App → Azure Runtime (Plan + Execute) →
Tool Registry (Discover) → RBAC/ABAC (Authorize) →
MCP Tool (Snowflake Views Query / Speed Layer Query) →
Result Processing → LLM (Summarize/Reason) →
Guardrails → Audit Log → Response → User
```

Complementary Runtime Flow (Joule + ECA Hybrid)

```
User → Joule Assistant UI → SAP Joule Runtime →
Skills (SAP Transaction/Read) → SAP Backend →
[Bridge: MCP Tool Call to Azure] → Azure Runtime →
MCP Tool (ECA Snowflake Query) → ECA Data Product →
Result → Joule Response Composer → User

Note: Hybrid pattern for use cases requiring both SAP
transactions and ECA analytics in a single workflow.
```

Data Product Consumption Flow

```
Consumer (BI / ML / GenAI / Agent) → Serving Interface Selection:
  Semantic Model → Power BI DirectQuery (Interactive BI)
  SQL Access → Snowflake SQL (Ad-hoc queries)
  Data APIs → REST/GraphQL (Application integration)
  NRT/RT Feeds → Snowflake Streams (Real-time dashboards)

Each interface enforces: RBAC (Consumer Role) + ABAC (Data Classification)
+ Rate Limits + Audit Logging
```

18. Related Architecture Patterns

ECA & AI Enablement is one of **three complementary AI architecture patterns** at Intel. Each pattern is purpose-built for a distinct class of use cases and maintains its own runtime, identity model, and governance boundaries.

Pattern	Scope	Entry Point
SAP Joule	SAP transactions, navigation, analytics, SAP knowledge	Joule Assistant bar in SAP apps
ECA / Azure Copilot Studio (this document)	Enterprise data analytics, agentic workflows, non-SAP integrations	Copilot Agent UX (Teams / MS365)
iGPT Platform	General-purpose chat, custom assistants, marketplace, document Q&A	igpt.intel.com/chat

High-Level Integration Points

- **ECA → Joule**: SAP data replicated via SLT/CIF is queryable in ECA. Hybrid use cases (SAP approval + ECA analytics) leverage MCP bridge and A2A protocol.
- **ECA → iGPT**: No direct integration. General-purpose AI needs are served by iGPT as a standalone platform.
- **BDC Connect**: SAP Business Data Cloud enables zero-copy data sharing between Joule and ECA/Azure runtimes via Delta Sharing (Databricks) and Snowflake Data Sharing.

Selection Guidance

For detailed use case routing, decision frameworks, and guidance on choosing between the three patterns, refer to:

> 📄 **AI Architecture Selection Guide** — `AI-Architecture-Selection-Guide.md`

Related Documents

Document	Location
SAP Joule Architecture Overview	`Joule Architecture/SAP-Joule-AI-Architecture-Overview.md`
iGPT Architecture Overview	`iGPT Architecture/iGPT-AI-Architecture-Overview.md`
AI Architecture Selection Guide	`AI-Architecture-Selection-Guide.md`

Appendix A: Component Glossary

Abbreviation	Full Name	Description
ABAC	Attribute-Based Access Control	Fine-grained access control using attributes
ADF	Azure Data Factory	ETL/ELT pipeline orchestration service
ADLS	Azure Data Lake Storage	Scalable object storage for data lakes
AGS	Access Group Service	Intel's enterprise entitlements service for group-based permissions
B/S/G	Bronze/Silver/Gold	Medallion architecture data layers
CDC	Change Data Capture	Technique for tracking data changes
CIF	Core Ingestion Framework	Controls data landing patterns
DARC	Data Analytics Reporting Center	Power BI finance reporting solution
ECA	Enterprise Cloud Analytics	ADLS + Databricks + Snowflake platform
HITL	Human-in-the-Loop	Approval process requiring human intervention
iGPT	Intel Generative Pre-trained Transformer	Intel's internal enterprise GenAI platform (igpt.intel.com) for chat, assistants, and marketplace
MCP	Model Context Protocol	Standard for tool/workflow integration
NRT	Near Real-Time	Sub-minute data latency
RAG	Retrieval-Augmented Generation	LLM grounding with enterprise context
RAGe	Retrieval Augmented Generation (enterprise)	SAP-specific enterprise RAG for Joule document grounding
RBAC	Role-Based Access Control	Permission management by role
RT	Real-Time	Sub-second data latency
SLT	SAP Landscape Transformation	SAP replication service
SoD	Segregation of Duties	Control preventing conflict of interest

Document Control

Version	Date	Author	Changes
1.0	February 2026	Sajiv Francis, Don Meyers	Initial release
1.1	March 2026	Sajiv Francis	Added Section 2: AI Architecture Patterns & Routing Guide (Joule, ECA/Azure, iGPT three-pattern model); updated Runtime Routing to include iGPT; expanded glossary
1.2	March 2026	Sajiv Francis	Restructured Section 2: moved detailed use case routing, capability patterns, and interoperability content to standalone AI Architecture Selection Guide; retained high-level pattern comparison and references

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End of Document